Fatemehsadat Mousavinasab

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LinkedIn | Portfolio | Google Scholar
U.S. Permanent Resident (no sponsorship required) | English (fluent), Persian (native)

SUMMARY

Immunologist and Ph.D. candidate in Biomedical Sciences (expected March 2027, GPA 4.19) with over 10 years of hands-on laboratory research and 10 published papers (one more under review). Able to run a research laboratory end to end: setting up new assays and animal models, managing daily lab operations, and writing SOPs to ISO/IEC 17025 standards. Experienced teacher who designs courses and teaches both theory and hands-on laboratory sessions, with 80–100 students and interns trained in immunology and molecular techniques. Research focuses on regulatory T cells in autoimmune (celiac disease) and allergic (eosinophilic esophagitis) conditions. Seeking research scientist and teaching (instructor) roles in academic, biotech, and pharmaceutical settings.

RESEARCH, TEACHING, AND LABORATORY EXPERIENCE

Ph.D. Candidate, Institute for Biomedical Sciences, Georgia State University *2022 – Present*

- **Developed a new humanized mouse model** to study peripheral regulatory T cells in oral tolerance to gluten in celiac disease.
- Designed and ran in vivo studies across different mouse groups.
- **Profiled gut T-cell populations by 10+ color flow cytometry** (activated CD4+ cells, Tregs, Th17/Treg balance, cytokines), histology, and functional assays.
- **Established hybridomas specific for human class II MHC/peptide complexes** (thymoma fusion, HLA-DQ8 dendritic cell co-culture, peptide testing).
- **Published a step-by-step protocol** for the hybridoma method (Bio-protocol, 2026) so other labs can reproduce it.
- **Managed a colony of 70–100 mice (12–14 cages)**, including injections, gavage, blood sampling, surgery, and organ isolation.
- Trained new students in core laboratory techniques.

Head of Immunology Section & Training Lead, GenIran Research Lab, Tehran, Iran *2020 – 2022*

- **Led the immunology section** and its immunoassay-based research projects.
- **Trained about 80–100 students and interns** through a hands-on immunoassay course, designing the syllabus and practical sessions.
- Mentored students in immunoassay and molecular genetic workflows (PCR, RT-qPCR), from sample preparation to data interpretation.
- **Wrote SOPs and quality records to ISO/IEC 17025 requirements** to prepare the laboratory for accreditation.

Researcher, Research Institute for Gastroenterology & Liver Diseases, Tehran, Iran *May – Dec 2019*

- Designed a research proposal and in vitro study on tofacitinib and M1/M2 macrophage polarization in infliximab non-responder ulcerative colitis, based on a review of IBD immunopathology and anti-TNF resistance.

Laboratory Supervisor, Shahid Beheshti University of Medical Sciences, Tehran, Iran *2017 – 2019*

- Taught practical immunology techniques and instrument operation to master's students.
- Mentored and advised master's students and lab researchers, and proctored student practical exams.

M.Sc. Researcher, Shahid Beheshti University of Medical Sciences, Tehran, Iran *2013 – 2016*

- Studied regulatory T cells in esophageal biopsies from patients with eosinophilic esophagitis, GERD, and controls.
- Processed and analyzed patient tissue specimens; **results published in Allergologia et Immunopathologia (2019)**.

Clinical Laboratory Rotation (B.Sc. Medical Laboratory Sciences), Valiasr Hospital, Arak, Iran *2012*

- Completed a 6-month full-time rotation through Hematology (Sysmex analyzers), Chemistry, Microbiology, Immunology, and Blood Bank, processing and analyzing patient specimens.

EDUCATION

Ph.D., Biomedical Sciences, Georgia State University, Atlanta, GA *Expected Mar 2027*

GPA 4.19. Dissertation: Peripheral regulatory T cells in oral tolerance to gluten in celiac disease mouse model. Mentor: Leszek Ignatowicz, Ph.D.

M.Sc., Medical Immunology, Shahid Beheshti University of Medical Sciences, Tehran, Iran *2013 – 2016*

GPA 3.86. Thesis: Regulatory T cells in esophageal tissue of patients with eosinophilic esophagitis vs. GERD and controls.

B.Sc., Medical Laboratory Sciences, Arak University of Medical Sciences, Iran *2008 – 2012*

GPA 3.93

TECHNICAL SKILLS

Immunology & cell biology: multicolor flow cytometry (FlowJo, compensation), FACS/MACS cell sorting, primary cell culture, dendritic cell generation, T-cell hybridoma generation and peptide testing

Molecular biology: DNA/RNA isolation, PCR, real-time qPCR, primer design, cDNA synthesis, gel electrophoresis

Protein & assays: ELISA, Western blot, SDS-PAGE, BCA, exosome isolation

Histology & imaging: tissue processing, H&E staining, immunohistochemistry, immunofluorescence, ImageJ

In vivo models: mouse colony management, disease induction, injections, gavage, surgery, organ isolation

Microbiology: Gram and capsule staining, biochemical identification tests, antibacterial susceptibility testing (disc and well diffusion)

Quality (QC/QA): ISO/IEC 17025 documentation and standards, instrument operation and training, analytical result interpretation

Software: GraphPad Prism, SPSS, REST (qPCR analysis), FlowJo, ImageJ, EndNote, NCBI/BLAST

PUBLICATIONS

Original articles

1. **Mousavinasab F**, Szurek EA, Kuczma MP, Cebula A, Murray JA, Marietta E, Ignatowicz L. Investigating the role of peripheral regulatory T cells in oral tolerance to gluten through the development of a novel mouse model of celiac disease. Frontiers in Immunology (under review).
2. **Mousavinasab F**, Szurek EA, Cebula A, Ignatowicz L, Kuczma MP. Improved protocol for establishing CD4+ hybridomas specific for human class II MHC/peptide complex. Bio-protocol. 2026;16(13):e5737. doi:10.21769/BioProtoc.5737.
3. Khatami S, Ebrahimi S, Mahjoub FE, Yousefi A, Hosseini AH, Khoshmirsafa M, **Mousavinasab F**, Mansouri M, Shabani M, Mesdaghi M. Detection of eosinophilic cell-free granules based on expression of CCR3 and MBP markers in colonic biopsy of pediatric patients with suspected and confirmed eosinophilic colitis. Iran J Immunol. 2026;23(2). doi:10.22034/iji.2026.109789.3152.
4. Ebrahimi S, Khatami S, Mahjoub FE, Monajemzadeh M, Yousefi A, Hosseini A, Khoshmirsafa M, **Mousavinasab F**, Mansouri M, Shabani M, Mesdaghi M. Detection of eosinophilic cell-free granules based on expression of CCR3 and MBP markers in esophageal biopsy specimens of patients with suspected and confirmed eosinophilic esophagitis. Pediatr Gastroenterol Hepatol Nutr. 2025;28(1):19-26. PMID: 39839467.
5. **Mousavinasab F**, Babaie D, Nilipour Y, Mansouri M, Imanzadeh F, Dara N, Rohani P, Khatami K, Sayyari A, Khoddami M, Kazemiaghdam M, Mesdaghi M. Increased number of regulatory T cells in esophageal tissue of patients with eosinophilic esophagitis in comparison to gastroesophageal reflux disease and control groups. Allergol Immunopathol. 2019;47(5):431-436.
6. Abdolahy M, Rasouli Sh, Babaie D, Dara N, Imanzadeh F, Sayyari A, Rouhani P, Khatami K, Kazemiaghdam M, Nilipour Y, Khoddami M, Ghadimi F, **Mousavinasab F**, Mesdaghi M. Increased regulatory T cells in peripheral blood of children with eosinophilic esophagitis. Gastroenterol Hepatol Bed Bench. 2021;25-30.

Reviews and book chapter

7. Aghaei-Zarch SM, Nia AHS, Nouri M, **Mousavinasab F**, et al. The impact of particulate matters on apoptosis in various organs: mechanistic and therapeutic perspectives. Biomed Pharmacother. 2023;165:115054. PMID: 37379642.
8. Hoorzad P, **Mousavinasab F**, Tofigh P, et al. Understanding the lncRNA/miRNA-NFκB regulatory network in diabetes mellitus: from function to clinical translation. Diabetes Res Clin Pract. 2023;202:110804. PMID: 37369279.
9. **Mousavinasab F**, Karimi R, Taheri S, et al. Microbiome modulation in inflammatory diseases: progress to microbiome genetic engineering. Cancer Cell Int. 2023;23:271.
10. **Mousavinasab F**, Razani E. Harnessing the potential of vesicular stomatitis virus as a novel therapeutic strategy for cancer treatment. Iran J Blood Cancer. 2023;15(2):117-127.
11. Salehi-Sardoei A, **Mousavinasab F**, Sayyed RZ, et al. Endophytic fungi: perspectives for microbial engineering (Ch. 11). In: Biostimulants and Protective Biochemical Agents. Elsevier; 2024:167-220. (Book chapter)

HONORS AND AWARDS

- **Dr. Suresh Sitaraman Award** (top three abstracts), Digestive Disease Research Day, Emory–Georgia State University (Sep 2026).
- **Poster Presentation Award**, AAI Immunology 2026, Boston, MA (Apr 2026).
- **Best Oral Presentation**, Basic and Advanced Translational Immunology Course, Jaipur, India (Oct 2019).
- **Travel grants:** FIMSA (Jaipur, 2019), EFIS (SEEIS 2018, Yerevan, Armenia), EAACI junior member (Helsinki, 2017).

PRESENTATIONS

- Digestive Disease Research Day, Emory–Georgia State University (Sep 2026). Oral presentation on peripheral regulatory T cells in oral tolerance to gluten.
- AAI Immunology 2026, Boston, MA (Apr 2026). Poster presentation on peripheral regulatory T cells in celiac disease.
- 4th Annual Graduate Conference for Research, Scholarship, and Creative Activity, Atlanta, GA (Feb 2026). Poster: elevated autoreactive CD4+Foxp3- T-cell clones in humanized HLA-DQ8 mice, revealed by Nur77-GFP T-cell hybridomas.
- 2025 Southeast Regional Clinical & Translational Science Conference, Pine Mountain, GA (Mar 2025). Presentation on creating a novel mouse model of celiac disease.
- 3rd Annual Graduate Conference for Research, Scholarship, and Creative Activity, Atlanta, GA (Feb 2025). Presentation on the novel mouse model of celiac disease.
- 2nd Research Day, Institute for Biomedical Sciences, Georgia State University (2025). Oral presentation on the murine model of gluten-specific oral tolerance.

SERVICE AND MEMBERSHIPS

- Vice President, Translational Biomedical Sciences Student Association, Georgia State University (2024 – Present). Lead the planning of scientific and social events for graduate students in the Institute for Biomedical Sciences, including organizing the annual Research Day.

- Judge, Georgia State Undergraduate Research Conference (April 2024 and April 2025).
- Peer reviewer, Cellular, Molecular and Biomedical Reports (2023): six manuscripts, listed on Web of Science.
- Member: American Association of Immunologists (AAI); International Union of Immunological Societies (IUIS); European Academy of Allergy and Clinical Immunology (EAACI); Women Speakers, FIMSA.

RESEARCH INTERESTS

Autoimmunity and tolerance (thymic and peripheral regulatory T cells, tolerance to self and dietary antigens); immunoregulation in IBD and GI disorders; macrophage polarization; allergy and cancer immunology (immune profiling, NK cells and exosomes in hematologic malignancies).

REFERENCES

Leszek Ignatowicz, Ph.D., Distinguished University Professor, Center for Translational Immunology, Georgia State University | lignatowicz@gsu.edu | 404-413-6685

Cynthia Nau Cornelissen, Ph.D., Distinguished University Professor; Director, Center for Translational Immunology; Associate Director, Institute for Biomedical Sciences, Georgia State University | ccornelissen@gsu.edu | 404-413-6680

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